STRENGTH OF MATERIALS

Subject Code : 23A01302T

UNIT – I

Simple Stresses and Strains

Handwritten Study Notes

Syllabus topics covered in this unit :

Elasticity and plasticity — Types of stresses and strains — Hooke's law — Factor of safety, Poisson's ratio — Relationship between elastic constants — Bars of varying section — Stresses in composite bars

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→ 1.1 Introduction – Strength of Materials
→ 1.2 Stress–Strain Curve (Mild Steel)
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1.1 Introduction – Strength of Materials

Strength of Materials (Mechanics of Solids) studies how a body responds to external loads – the **internal resistance** (stress) and the **deformation** (strain) it develops.

- Rigid body mechanics: body is assumed non-deformable (statics/dynamics).
- Deformable body mechanics: loads cause change of size/shape – our subject.
- Aim of design : stresses < permissible limit, deformations < allowable limit.

Elasticity and Plasticity

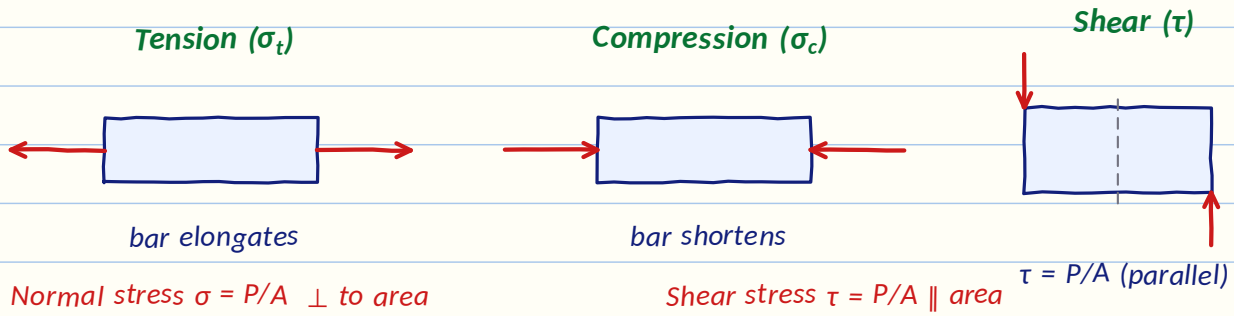
- **Elasticity** : property by which a body regains its original size and shape on removal of the load. Perfectly elastic – quartz, steel (within limit).
- **Plasticity** : body does not regain its shape – permanent (residual) deformation remains. e.g. lead, clay, steel beyond the yield point.
- Elastic limit : max stress up to which the body is perfectly elastic.
- Ductile : large plastic strain before failure (mild steel, copper). Brittle : fails with little strain (cast iron, glass, concrete).
- Malleability – can be hammered into sheets ; Toughness – energy absorbed up to fracture.

Stress

Internal resistance offered per unit area to the external load.

$$\sigma = \frac{P}{A} \qquad (N/mm^2 = MPa)$$

- $1\ N/mm^2 = 1\ MPa = 10^6\ N/m^2\ (Pa)$; $1\ GPa = 10^3\ MPa = 10^9\ Pa$.



Normal stress $\sigma = P/A$ $\perp$ to area

Shear stress $\tau = P/A$ $\parallel$ area

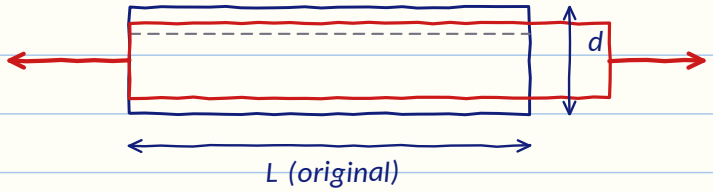
Types of Stresses

Type	Cause	Formula
Tensile	Pulling force	$\sigma_t = P/A$
Compressive	Pushing force	$\sigma_c = P/A$
Shear	Forces ∥ to area	$\tau = P/A$
Bearing	Contact pressure	$\sigma_b = P/(d \cdot t)$
Thermal	Restrained expansion	$\sigma = \alpha T E$
Volumetric (bulk)	Hydrostatic p	$\sigma = p$

Strain

Ratio of change in dimension to original dimension – dimensionless.

- Linear (longitudinal) strain $\varepsilon = \delta L/L$ (δL = change in length)
- Lateral strain = $\delta d/d$ – opposite in sign to longitudinal strain.
- Shear strain φ (γ) = angular distortion in radians = $\tan \varphi \approx \varphi = \delta/h$
- Volumetric strain $\varepsilon_v = \delta V/V = \varepsilon_x + \varepsilon_y + \varepsilon_z$



Longitudinal strain $\varepsilon = \delta L/L$ (+)

Lateral strain = $-\delta d/d$ (–)

Poisson's ratio μ = – lateral/longitudinal

μ = 0.25 – 0.33 for metals (max 0.5)

Hooke's Law

Within the elastic (proportional) limit, stress is directly proportional to strain.

$$\sigma \propto \varepsilon \Rightarrow \sigma = E\,\varepsilon \Rightarrow E = \frac{\sigma}{\varepsilon}$$

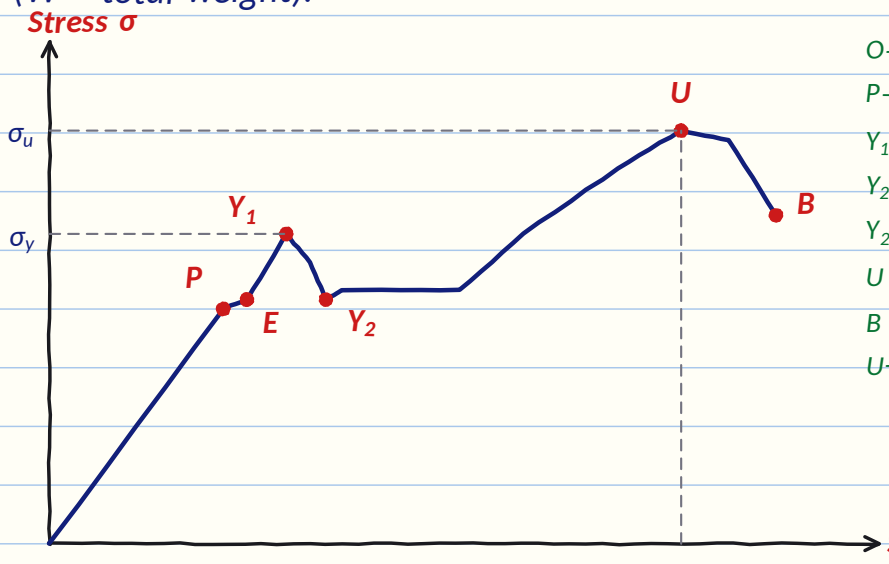
- E = Young's modulus (modulus of elasticity) – tensile/compressive.
- G (C, N) = Modulus of rigidity = τ/φ (shear).
- K = Bulk modulus = $p/(\delta V/V)$ (volumetric).

Material	E (GPa)	G (GPa)	μ
Mild steel	200–210	80	0.25–0.30
Cast iron	100–120	40	0.23–0.27
Copper	100–110	40	0.32–0.35
Aluminium	70	26	0.33

Deformation of a Bar under Axial Load

$$\delta L = \frac{P\,L}{A\,E}$$

- E A is the "axial rigidity" – larger E A means smaller elongation.
- Bar under its own weight (length L, weight density w): $\delta L = (w\,L^2)/2E = W\,L/2\,A\,E$ (W = total weight).



O–P : proportional limit
P–E : elastic limit
Y_1 : upper yield pt
Y_2 : lower yield pt
Y_2–: plastic flow
U : ultimate stress
B : breaking (fracture)
U–B : necking

1.2 Stress–Strain Curve (Mild Steel)

- Obtained by Universal Testing Machine (UTM) on a standard specimen.
- O–P : straight line – Hooke's law valid ; slope = E.
- P–E : slightly beyond P – still elastic but not proportional.
- Y_1 , Y_2 : yield – strain increases with almost no increase in load. Material starts to flow plastically. Upper yield Y_1 is sensitive ; use lower yield Y_2.
- $Y_2 \rightarrow U$: strain hardening – stress rises again up to ultimate stress.
- $U \rightarrow B$: necking – area reduces locally, load drops, specimen breaks at B.

Important results from the test

- Percentage elongation = $(L_f - L_0)/(L_0) \times 100$ (ductility measure)
- Percentage reduction in area = $(A_0 - A_f)/(A_0) \times 100$
- True stress = P / (instantaneous area) > nominal stress after yield.
- Proof stress : for materials with no definite yield point (aluminium, copper) take stress for 0.2 % permanent strain (draw a line ∥ to OP from 0.2 % strain).
- Cast iron (brittle): no yield point, fails suddenly, strong in compression.

Factor of Safety (F.O.S.)

$$F.O.S. = \frac{\text{Ultimate stress}}{\text{Working (permissible) stress}}$$

- For ductile materials, base on yield stress : F.O.S. = $\sigma_y / \sigma_{working}$.
- Typical values: steel 1.85–3 ; cast iron 4–6 ; timber 7–10 ; concrete 3–4.
- Margin of safety = F.O.S. – 1.
- Why? uncertainty in loads, material defects, stress concentration, workmanship.

Example 1

A steel rod 20 mm dia., 2 m long carries a tensile load 60 kN. $E = 2\times10^5$ N/mm^2. Find stress, strain and elongation.

$A = \pi/4 \times 20^2 = 314.16$ mm^2

$\sigma = P/A = 60000 / 314.16 =$ **190.99 N/mm^2**

$\varepsilon = \sigma/E = 190.99 / (2\times10^5) =$ **9.55×10^{-4}**

$\delta L = \varepsilon L = 9.55\times10^{-4} \times 2000 =$ **1.91 mm**

Example 2 (Factor of safety)

A bar of 25 mm dia. fails at 180 kN. If the working load is 45 kN, F.O.S. = ?

Ultimate load / working load = 180/45 = **4**

1.3 Poisson's Ratio and Elastic Constants

Poisson's ratio μ (or 1/m) = Lateral strain/Longitudinal strain (within elastic limit).

- Axial pull ⇒ length ↑, lateral dimensions ↓ (and vice versa). Value 0.25 to 0.33 for most metals; theoretical max 0.5.

Three-dimensional stress system

Strain in x-direction when σ_x, σ_y, σ_z (tension +) act:

$$\varepsilon_x = \frac{\sigma_x}{E} - \mu\frac{\sigma_y}{E} - \mu\frac{\sigma_z}{E}$$

- Similarly for ε_y and ε_z.
- Volumetric strain $\varepsilon_v = \varepsilon_x+\varepsilon_y+\varepsilon_z = (\sigma_x+\sigma_y+\sigma_z)/E(1-2\mu)$
- For a bar loaded in x-direction only : $\varepsilon_v = \varepsilon(1-2\mu)$; $\delta V = \varepsilon(1-2\mu)V$.
- Since $\varepsilon_v \geq 0 \Rightarrow \mu \leq 0.5$.

Relationship between E, G, K and μ

$$E = 2G(1+\mu)$$

$$E = 3K(1-2\mu)$$

$$E = \frac{9KG}{3K+G} \quad ; \quad \mu = \frac{3K-2G}{2(3K+G)}$$

Derivation of E = 2G(1+μ) (pure shear)

- Take a square element ABCD under shear stress τ on all four faces.
- Shear produces a diagonal tension τ and diagonal compression τ at 45° (principal stresses +τ and −τ).
- Strain of diagonal BD = $\tau/E + \mu\,\tau/E = \tau/E(1+\mu)$.
- From geometry, diagonal strain = ½ (shear strain φ) = $\tau/2G$.
- Equate: $\tau/2G = \tau/E(1+\mu) \Rightarrow$ **$E = 2G(1+\mu)$**

Derivation of E = 3K(1−2μ) (cube under p)

- Cube subjected to uniform pressure p on all faces: $\sigma_x=\sigma_y=\sigma_z = p$.
- Linear strain $\varepsilon = p/E(1-2\mu)$, volumetric strain $\varepsilon_v = 3\varepsilon = 3p/E(1-2\mu)$.
- By definition $K = p/\varepsilon_v \Rightarrow$ **$E = 3K(1-2\mu)$**
- Eliminating μ from the two results gives $E = 9KG/(3K+G)$.

Example 3

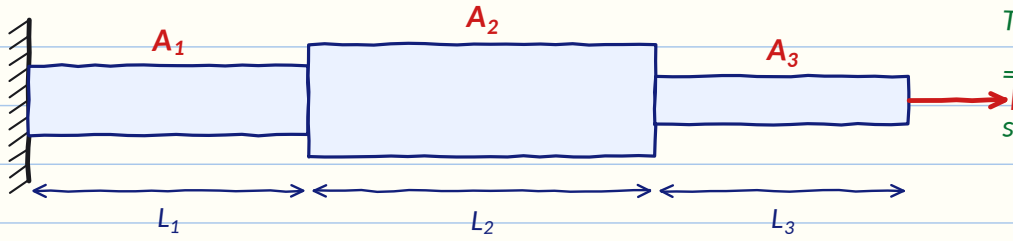
For a material $E = 2.1\times10^5$ N/mm², $G = 0.8\times10^5$ N/mm². Find μ and K.

$\mu = E/2G - 1 = 2.1/(2\times0.8) - 1 =$ **0.3125**

$K = E/[3(1-2\mu)] = 2.1\times10^5/[3(1-0.625)] =$ **1.87×10^5 N/mm²**

1.4 Bars of Varying Section

Load P acts on a bar whose area changes in steps. Each portion carries the same internal force (if loads act only at the ends).



Total $\Delta L = \Delta L_1 + \Delta L_2 + \Delta L$

$= \sum P L / (A E)$

same $P \Rightarrow \sigma_n = P/A_n$

$$\Delta L = \frac{P}{E}\left(\frac{L_1}{A_1} + \frac{L_2}{A_2} + \frac{L_3}{A_3}\right)$$

- If different E for each part: $\Delta L = P \sum (L_n)/(A_n E_n)$.
- Stress in each part is different: $\sigma_n = P/A_n$ (max stress in the smallest area).

Bar with several axial loads

- Draw a free-body diagram, split the bar at every load point.
- Find the axial force in each segment (tension +, compression –) starting from the free end.
- Total elongation = algebraic sum of the changes of each segment.

Example 4

A stepped bar: AB = 400 mm (A = 600 mm^2), BC = 300 mm (A = 400 mm^2), CD = 200 mm (A = 200 mm^2); axial pull 20 kN. E = 2×10^5 N/mm^2. Find total elongation.

$\Delta L = P/E(400/600 + 300/400 + 200/200)$

$= 20000/(2\times10^5)\ (0.667 + 0.75 + 1.0)$

$= 0.1 \times 2.417 =$ **0.2417 mm**

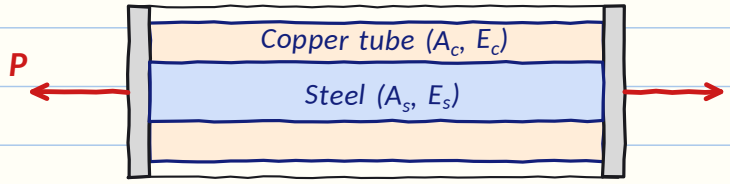
Tapering bar (circular, diameters $d_1 > d_2$)

$$\delta L = \frac{4 P L}{\pi E d_1 d_2}$$

- Rectangular taper (thickness t, width $b_1 \to b_2$): $\delta L = P L/(E t (b_1 - b_2)) \ln (b_1)/(b_2)$.

Temperature stresses

- A free bar heated by t°C expands $\delta L = \alpha L t$ – no stress.
- If expansion is prevented : $\sigma = \alpha t E$ (compressive on heating).
- Thermal load $P = \alpha t E A$. If support yields by δ: $\sigma = E(\alpha t - \delta/L)$.



Equilibrium : $P = P_s + P_c$

Compatibility : $\delta_s = \delta_c \Rightarrow \varepsilon_s = \varepsilon_c$

$\sigma_s / E_s = \sigma_c / E_c \Rightarrow \sigma_s = (E_s/E_c)\ \sigma_c$

$\sigma_s = m\ \sigma_c$, $m = E_s/E_c$ (modular ratio)

1.5 Composite (Compound) Bars

Two or more materials rigidly joined so that they deform together under load.

- Equilibrium : $P = P_1 + P_2 = \sigma_1 A_1 + \sigma_2 A_2$
- Compatibility : equal strain (same elongation) : $\varepsilon_1 = \varepsilon_2$

$$\frac{\sigma_1}{E_1} = \frac{\sigma_2}{E_2} \Rightarrow \sigma_1 = \frac{E_1}{E_2}\sigma_2$$

- Modular ratio $m = E_1/E_2$ (stiffer / softer).
- Load shared : $P_1 = P\,(A_1 E_1)/(A_{1E1} + A_{2E2})$, $P_2 = P\,(A_2 E_2)/(A_{1E1} + A_{2E2})$
- Equivalent area (in material 2): $A_{eq} = A_2 + m A_1$.

Example 5

Steel rod 20 mm dia. fits inside a copper tube 30 mm ext., 20 mm int. dia. Load 100 kN. $E_s = 2\times10^5$, $E_c = 1\times10^5$ N/mm^2. Find stresses.

$A_s = \pi/4(20)^2 = 314.2$ mm^2 ; $A_c = \pi/4(30^2 - 20^2) = 392.7$ mm^2

$\sigma_s = 2\,\sigma_c$ $(E_s/E_c = 2)$

$P = \sigma_s A_s + \sigma_c A_c = \sigma_c\,(2\times314.2 + 392.7) = 1021.1\,\sigma_c$

$\sigma_c = 100000/1021.1 =$ **97.93 N/mm^2**

$\sigma_s =$ **195.86 N/mm^2**

Temperature stresses in composite bars

- Different α : on heating, the material with higher α is restrained by the other ⇒ it is in compression, the other in tension.
- Equilibrium (no external load): $\sigma_1 A_1 = \sigma_2 A_2$ (equal & opposite forces).
- Compatibility: free expansion difference = elastic strains:

$$(\alpha_1 - \alpha_2)\,t = \frac{\sigma_1}{E_1} + \frac{\sigma_2}{E_2}$$

1.6 Shear, Bearing and Punching Stresses

- Shear stress acts **parallel** to the cross-section : $\tau = P/(A_{shear})$.
- Complementary shear : a shear stress on a plane is always accompanied by an equal shear stress of opposite sense on the perpendicular plane : $\tau_{xy} = \tau_{yx}$.
- Shear modulus $G = \tau/\varphi$ (φ = shear strain in radian).

Case	Shear area	τ
Single shear (rivet/bolt)	$\pi d^2/4$	$P / (\pi d^2/4)$
Double shear (clevis)	$2 \times \pi d^2/4$	$P / (2 \pi d^2/4)$
Punching a hole	$\pi d t$	$P / (\pi d t)$
Bearing (crushing)	$d \times t$	$P / (d t)$

Example 6 (Punching)

Force needed to punch a 25 mm hole in a 10 mm thick plate, shear strength τ_u = 300 N/mm^2.

Shear area = $\pi d t = \pi \times 25 \times 10 = 785.4$ mm^2

$P = \tau_u \times$ area = $300 \times 785.4 = 235\,619$ N ≈ **235.6 kN**

Example 7 (Double shear)

A 20 mm bolt in double shear carries 50 kN.

$\tau = P / (2A) = 50\,000 / (2 \times 314.16) =$ **79.58 N/mm^2**

1.7 Statically Indeterminate Bars and Temperature

Bar with intermediate loads (Example 8)

Uniform bar ABCD ($A = 500$ mm^2, $E = 2\times10^5$ N/mm^2) fixed at A. AB = 600, BC = 800, CD = 500 mm. Axial loads : 20 kN pull at D, 50 kN pull at C, 30 kN push at B.

Axial forces : CD = +20 kN ; BC = 20 + 50 = +70 kN ; AB = 70 – 30 = +40 kN

$\Delta L = 1/AE\ (20\times500 + 70\times800 + 40\times600)$ kN·mm $= (9\times10^7)/(500 \times 2\times10^5)$

$\Delta L =$ **0.9 mm** (elongation)

Bar fixed at both ends, load P at C (a from A, b from B)

- Equilibrium : $R_A + R_B = P$. Compatibility : total change of length = 0.
- AC extends under R_A ; CB shortens under R_B : $(R_A\ a)/AE = (R_B\ b)/AE$

$$R_A = \frac{P\ b}{L} \qquad R_B = \frac{P\ a}{L}$$

Example 9

Bar of 1.2 m between rigid walls, 60 kN axial load at 0.4 m from A ($a = 0.4$, $b = 0.8$).

$R_A = 60 \times 0.8 / 1.2 =$ **40 kN** ; $R_B = 60 \times 0.4 / 1.2 =$ **20 kN**

Example 10 (Thermal – rail)

12 m steel rail laid at 15°C, max temperature 45°C. $\alpha = 12\times10^{-6}$ /°C, $E = 2\times10^5$ N/mm^2.

Free expansion = $\alpha L t = 12\times10^{-6} \times 12000 \times 30 =$ **4.32 mm**

If ends fully restrained : $\sigma = \alpha\, t\, E = 12\times10^{-6} \times 30 \times 2\times10^{5} =$ **72 N/mm^2**

If 3 mm gap is left : $\sigma = E\,(\alpha\, t - \delta/L) = 2\times10^{5}(3.6\times10^{-4} - 2.5\times10^{-4}) =$ **22 N/mm^2**

1.8 More Solved Problems

Example 11 (Volumetric strain)

Bar 50 mm × 50 mm × 1000 mm, axial pull 100 kN, $E = 2\times10^5$ N/mm², $\mu = 0.3$. Find change in length and volume.

$\sigma = 100\,000 / 2500 = 40$ N/mm² ; $\varepsilon = 40 / 2\times10^5 = 2\times10^{-4}$

$\delta L = \varepsilon L =$ **0.2 mm**

$\varepsilon_v = \varepsilon (1 - 2\mu) = 2\times10^{-4} \times 0.4 = 8\times10^{-5}$

$\delta V = \varepsilon_v \times V = 8\times10^{-5} \times 2.5\times10^6 =$ **200 mm³**

Example 12 (Tapering circular bar)

Bar tapers uniformly from 40 mm to 20 mm dia. over 500 mm. P = 50 kN, $E = 2\times10^5$ N/mm².

$\delta L = 4 P L/(\pi E d_1 d_2) = (4 \times 50000 \times 500)/(\pi \times 2\times10^5 \times 40 \times 20)$

$\delta L = 10^8 / 5.027\times10^8 =$ **0.199 mm**

Example 13 (Composite bar with temperature)

Steel ($A = 500$ mm²) and copper ($A = 1000$ mm²) bars rigidly joined, heated by 60°C. $\alpha_s = 12\times10^{-6}$, $\alpha_c = 17\times10^{-6}$, $E_s = 2\times10^5$, $E_c = 1\times10^5$ N/mm².

Equilibrium : $\sigma_s A_s = \sigma_c A_c \Rightarrow \sigma_s = 2\,\sigma_c$

Compatibility : $(\sigma_s)/(E_s) + (\sigma_c)/(E_c) = (\alpha_c - \alpha_s)\, t = 5\times10^{-6} \times 60 = 3\times10^{-4}$

$\sigma_c\,(1\times10^{-5} + 1\times10^{-5}) = 3\times10^{-4} \Rightarrow \sigma_c =$ **15 N/mm²** (compression)

$\sigma_s =$ **30 N/mm²** (tension) ; check : 30×500 = 15×1000 = 15 kN ✓

Example 14 (Self-weight)

Vertical steel rod 30 m long, density 78.5 kN/m³, $E = 2\times10^5$ N/mm² hangs freely.

Max stress at top = $w L = 78.5\times10^{-6}$ N/mm³ × 30000 mm = **2.355 N/mm²**

$\delta L = (w L^2)/2E = (78.5\times10^{-6} \times 30000^2)/(2 \times 2\times10^5) =$ **0.177 mm**

1.9 Important Short Questions (Exam Oriented)

- Q1. State Hooke's law. A: $\sigma \propto \varepsilon$ within elastic limit; $\sigma = E\varepsilon$.
- Q2. Define Poisson's ratio. A: ratio of lateral strain to longitudinal strain; 0.25–0.33 for metals.
- Q3. What is factor of safety? A: ultimate (yield) stress / working stress.
- Q4. Write the relation among E, G, K. A: $E = 2G(1+\mu) = 3K(1-2\mu) = 9KG/(3K+G)$.
- Q5. What is modular ratio? A: $m = E_1 / E_2$ used for composite bars.
- Q6. Define proof stress. A: stress producing a specified permanent strain (0.2 %) in materials with no clear yield point.

- Q7. What is complementary shear stress? A: equal & opposite shear stress on perpendicular planes.
- Q8. Why are temperature stresses zero in a freely expanding bar? A: no restraint ⇒ no strain energy / no elastic strain.

Quick revision

- $\sigma = P/A$; $\varepsilon = \delta L/L$; $E = \sigma/\varepsilon$; $\delta L = PL/AE$
- $E = 2G(1+\mu) = 3K(1-2\mu) = 9KG/(3K+G)$
- Composite bar : same ε ; $\sigma_1/E_1 = \sigma_2/E_2$; $\sum P = P$
- Thermal : $\sigma = \alpha\, t\, E$ (restrained)